

Dell Latitude 7490 OpenCore Configuration

Best Experienced With: Mac OS Sonoma or Ventura



First and foremost, this project wouldn't be here if it wasn't for CllloverLeafBG's work shown here:

<https://github.com/CloverLeafBG/Dell-Latitude-7490-OC-Hackintosh>

My project builds on the previous author's work, and has been adapted to work with Mac OS Sonoma. This project contains AI generated code.

The following boot arguments are all that is needed for me, your experience may vary:

alcid=13 watchdog=0 brcmf-dm-delay=300

To use DEBUG AirportBrcmFixup.kext add these boot-args:

-brcmfxdbg -liludbgall

System Configuration

- CPU: Intel Core i7-8650U
- iGPU: Intel UHD Graphics 620
- RAM: 32GB DDR4 2400Mhz
- SSD: WD Blue SN5000 2TB
- WiFi: BCM94360NG
- Display: 1920*1080 FullHD IPS
- Sound Card: Realtek ALC256
- LAN: Intel I219-LM
- Touchpad: Alps touchpad

BCM94360NG NSS:1 fix (AirportBrcmFixup.kext v. 2.1.9 mod)

This is probably the most meaningful contribution. The BCM94360NG problem was that on a cold macOS boot, the Broadcom driver applied a **single-transmit-chain constraint**, leaving the card at **NSS:1 instead of NSS:2**.

The fix was found by comparing the good NSS:2 state with the bad NSS:1 state (good state initiated by booting into Windows first), then instrumenting and reverse-engineering Apple's Broadcom driver. That led to the internal function `_wlc_stf_txchain_set`. Then discovered that it takes four arguments, and logging showed the bad path was specifically setting:

constraint/reason 2 to a chain value of **0x1**, whereas the working state used **0x3**.

The custom AirportBrcmFixup patch hooks that function and changes only that specific bad case:

```
reason == 2 && txchain == 1 → change the chain mask to 0x3
```

Then it lets Apple's original function continue normally.

So the fix **doesn't artificially report NSS:2 or generally force Wi-Fi settings**. It prevents Apple's driver from incorrectly disabling the second transmit chain in that one situation. The result is that the BCM94360NG comes up with both TX chains enabled and maintains **NSS:2 without needing the Windows-boot workaround**.

Note: For MacOS Sonoma you must use OpenCore Legacy Patcher in order to get BCM94360NG card functioning, as Sonoma removed driver support for this card. Latest version was used successfully.

<https://github.com/dortania/Opencore-Legacy-Patcher>

Sleep fix for AlpsHID.kext

My particular Latitude was failing to fall asleep because the cursor would keep moving slightly, as soon as the lid was closed, thus preventing sleep from occurring. The modified AlpsHID.kext specifically disables cursor input once lid is closed, and resumes functioning when lid is opened again. Using this kext there has been no issue with falling asleep. Previously it was recommended to disable USB PowerShare and USB Wake Support to overcome this specific issue, and it may not be needed anymore if your trackpad can use this kext.

Confirmed Working:

USB ports (USBA and USBC)
HDMI
Sound
Brightness and Volume adjust keys function properly
Sleep works
Airplay (extensively tested)
Trackpoint AKA "Pointing Stick"
Filevault
Ethernet
Low Power Mode
USB PowerShare (Option in bios may be turned on without affecting sleep)
Graphics Acceleration
Internal Microphone
Internal Webcam
MicroSD Card Reader
NSS:2 for BCM94360NG

Have Not Tested:

Airdrop, Messages compatibility, Smart Card Reader, Fingerprint sensor

Clear unknown BIOS password: Just enter the System Number displayed on BIOS unlock screen into this website:

<https://bios-pw.org/>.

Video Tutorial: https://www.youtube.com/watch?v=0HTvba_bN64